

The Effect of Cardiac Rehabilitation in Paediatric Fontan Circulation Patients: A Prospective Intervention Study.

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Background and Objectives: The Fontan procedure, a palliative surgery for univentricular heart physiology, often reduces exercise capacity and quality of life. This study aimed to evaluate the impact of cardiac rehabilitation (CR) on improving outcomes in Fontan patients to inform evidence-based care. **Materials and Methods:** Fontan patients aged 8-30 participated in a structured CR program for at least three months. The program included weekly aerobic and resistance training sessions with educational and nutritional guidance. Baseline and post-CR assessments included cardiac function, fitness, daily activity, and health-related quality of life (HRQOL). **Results:** The cohort included ten Fontan patients, of whom six had a right systemic ventricle. CR significantly improved cardiorespiratory fitness, as seen in VO2 max (from 27.92 ± 5.15 to 34.69 ± 1.14 mL/kg/min, $p = 0.0089$) and percent predicted VO2 (from 0.67 ± 0.18 to 0.90 ± 0.02 , $p = 0.005$). VCO2 increased by $+8.68 \pm 8.59$ mL/kg/min but did not reach statistical significance ($p = 0.05$). Most haemodynamic and ventilatory parameters showed no significant improvement. All the SF-36 questionnaire domains showed significant HRQOL gains ($p < 0.001$). High adherence (85-93%), no adverse events, and reduced NT-proBNP levels supported the program's safety. **Conclusions:** This study's findings have important implications for the care of Fontan patients. CR significantly improved exercise capacity and HRQOL in Fontan patients across various anatomies, particularly in the right systemic ventricle. Integrating physical conditioning into standard care could reduce long-term morbidity and mortality; however, further research is needed to refine the protocols and confirm sustained benefits.

Fenestration improves clinical outcome of the fontan procedure: a prospective, randomized study.

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The Fontan procedure is the definitive operation for palliation of complex congenital heart disease with single-ventricle physiology. Fenestration of the Fontan circuit allows for shunting of deoxygenated blood to the systemic circulation. This procedure improved the clinical outcomes of patients who are at high risk for poor Fontan results. However, it is controversial whether fenestration is beneficial for standard-risk patients. This prospective, randomized trial evaluated the clinical utility of fenestration in patients with standard preoperative risk profiles for Fontan surgery. Forty-nine consecutive patients were assigned to undergo either a fenestrated (25 patients) or nonfenestrated (24 patients) Fontan procedure. The fenestrated and nonfenestrated groups were comparable with respect to age ($P=0.944$), body surface area ($P=0.250$), number of preoperative risk factors for poor outcome ($P=0.681$), cardiopulmonary bypass time ($P=0.302$), number of patients who required aortic cross-clamping ($P=0.240$), preoperative oxygen saturation ($P=0.101$), and number of patients with dominant left ventricular morphology ($P=0.534$). Patients in the fenestrated group had 55% less total chest tube drainage ($P=0.036$), 41% shorter total hospitalization ($P=0.018$), and 67% fewer additional procedures in the postoperative period ($P=0.006$) than those in the nonfenestrated group. Baffle fenestration performed at the time of Fontan surgery improves short-term outcome in standard-risk patients by decreasing pleural drainage, hospital length of stay, and need for additional postoperative procedures.

Serial cardiopulmonary exercise testing in young patients after one-and-half ventricle repair and Fontan procedure: a comparative study.

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The Fontan procedure is a palliative surgical treatment for different congenital heart diseases with a univentricular heart, but it has been associated with decreased exercise capacity, cardiovascular morbidity, and premature mortality. The one-and-half ventricle repair (1.5VR) was introduced as an alternative to the Fontan procedure, specifically for selected patients with borderline hypoplastic right ventricle (HRV), aiming for a more physiological circulation. Despite these efforts, the benefit of 1.5VR over Fontan circulation comparison on clinical and functional outcomes remains unclear. The aim of this study was to investigate and compare young patients with HRV after 1.5VR with those with functional single right or left ventricles (FSRV or FSLV) after Fontan palliation over a 10-year follow-up period. In this retrospective observational study, serial cardiopulmonary exercise tests (CPETs) performed in patients with 1.5VR and Fontan circulation between September 2002 and March 2024 have been analysed. Only patients with at least 10 years of follow-up were considered. A total of 43 patients were included (age at baseline 8.6 ± 2.6 years): 21 with FSLV, 12 with FSRV, and 10 with 1.5VR. No differences in cardiorespiratory fitness and efficiency were shown at the first CPET assessment among the three groups. At 10-year follow-up, 1.5VR had higher cardiorespiratory fitness and efficiency compared to FSLV and FSRV patients. These findings suggest that the 1.5VR may provide superior long-term functional outcomes than the Fontan procedure in patients with borderline HRV. Further studies are needed to evaluate the impact on hard clinical endpoints.

Postoperative graft thrombosis in Fontan procedure.

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Fontan repair could be used as the definitive palliation in many forms of complex cyanotic congenital heart disease. But thrombosis can occur after a modified Fontan operation (right atrium-right ventricular connection with a conduit). Appropriate management of this complication includes thrombolytic therapy, thrombectomy and revision (if surgically remediable causes of the thrombosis are identified) or redo operation. Two repair operations were performed for the treatment of thrombosis of the right side of the heart in patients in whom we had previously performed Bjork modification (right atrium-right ventricular connection with a conduit). The thromboses occurred 6 and 9 years after the operation, respectively. In both cases, the redo Fontan operation was successfully performed using a dacron tube graft. Patients were anticoagulated after the operation. Risk of thrombosis of the right side of the heart after the Fontan repair may be minimized with the use of prophylactic anticoagulation in high-risk patients soon after the operation.

[Modified Fontan procedure for complex cardiac anomaly with nonconfluent pulmonary artery].

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Modified Fontan procedure was successfully performed in two patients with complex cardiac anomaly and nonconfluent pulmonary artery. A nonconfluent portion of the pulmonary artery was interposed with a

xenopericardial roll. In one patient, the central pulmonary artery was completely absent. Two-staged reconstruction of the central pulmonary artery was carried out through a left thoracotomy and a median sternotomy before the Fontan procedure. In the other patient, a juxtaductal segment of the pulmonary artery was nonconfluent. Correction of the nonconfluent pulmonary artery was performed concomitantly with the Fontan procedure. Fenestrated Fontan procedure seemed to be the beneficial modification for survival. Both patients have been day well during the midterm follow-up period.

Risk Factors for Mid-Term Liver Disease After the Fontan Procedure.

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The Fontan procedure is a palliative surgery performed for patients with complex congenital heart disease who exhibit functional single ventricular physiology. Although clinical outcomes of the Fontan procedure have improved in recent years and most patients who undergo the procedure reach adulthood, Fontan-associated liver disease (FALD) is a noncardiovascular complication that has become increasingly common; its risk factors remain unknown. A total of 95 patients who underwent the Fontan procedure and who were followed up for at least three years at Gunma Children's Medical Center and Kitasato University Hospital between 1996 and 2015 were retrospectively enrolled in this study. The mean age of the patients at the time of Fontan procedure was 2.3 ± 1.4 years. Overall, 21 patients (23.1%) experienced FALD. All Fontan procedures were performed with extracardiac total cavopulmonary connection using 16-mm expanded polytetrafluoroethylene grafts. The presence of systemic right ventricle, requirement of pulmonary vasodilator, application of a non-fenestrated Fontan procedure, and absence of fenestration flow at the time of follow-up catheter examination were identified as predictors of FALD using univariate analysis. All these factors, except the requirement of pulmonary vasodilator, remained significant predictors of FALD in multivariate logistic regression analysis. Patients with a systemic right ventricle who undergo the Fontan procedure are at a high risk of FALD in the mid-term. Creating fenestration at the time of Fontan and maintaining the fenestration flow may reduce the mid-term risk of FALD.

In Vitro Ultrasound Measurements of Powered and Unpowered Total Cavopulmonary Connection.

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Three-staged Fontan palliation is performed on children suffering from single ventricle congenital heart disease. The series of surgical procedures reroutes blood from the vena cavae directly to the pulmonary arteries, creating a total cavopulmonary connection (TCPC). A viscous impeller pump (VIP) is currently being developed as a cavopulmonary assist device that can modestly augment cavopulmonary flow, reduce systemic venous pressure, and improve ventricular preload. This study used ultrasound to visualize complex flow patterns in powered and unpowered in vitro mock Fontan circulations. The idealized TCPC was modeled with a silicone mold and blood analog made of water and glycerol that was seeded with 10- μ m glass beads. B-mode, color Doppler, and pulsed-wave Doppler images were used to visualize complex flow patterns in the idealized TCPC with (1) no VIP, (2) static VIP, and powered VIP rotation rates of (3) 500 and (4) 2,000 rotations per minute (RPM). Pulsed-wave Doppler data showed higher mean velocities and greater variance in the outlets relative to the larger inlets. The maximum inlet

velocity $\pm$ SD increased from 10.9 ± 3.53 cm/s with no VIP to 15.9 ± 1.03 when the VIP was rotating at 2,000 RPM. Likewise, the maximum outlet velocity increased from 14.9 ± 11.2 cm/s to 18.9 ± 7.25 cm/s at 2,000 RPM. The faster mean velocities with the VIP rotating suggest that the pump augments cavopulmonary flow. The results of this study suggest that measuring complex flow patterns with ultrasound in vivo could be used clinically to optimize VIP positioning and rotation rate during and after implantation.

Hemoptysis in a single lung Fontan: percutaneous approach.

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The Fontan procedure is sometimes the last palliative surgery for complex congenital heart disease with single-ventricle physiology. According to some authors, atrial-baffle fenestrations with right-to-left shunts have improved the surgical outcome of the Fontan operation. Percutaneous fenestration closure at a later stage may increase saturations and reduce the risk of embolic complications. A case could be made that the fenestration should be left patent if the patient, having lost the continuity of the pulmonary artery to one lung, only has a single functional lung. The case of an 11-year-old boy with a severe form of Ebstein anomaly is reported. He had undergone a fenestrated Fontan procedure and had lost the continuity to the left pulmonary artery. Further intervention was discouraged at other centers but he was eventually referred to our institution due to increasing cyanosis and severe hemoptysis. Several right-sided arteriovenous pulmonary malformations and collateral arteries were embolized percutaneously; a major right-to-left shunt at the fenestration in the atrial baffle was also occluded with an Amplatzer device. His saturation rose from 78 to 94% and his exercise tolerance increased, without further hemoptysis.

Modified Fontan procedure in patients with previous ascending aorta-pulmonary artery anastomosis.

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The Fontan procedure has proved useful in the surgical treatment of patients with tricuspid atresia, univentricular heart, and other complicated cardiac anomalies. Previous palliation of these patients with ascending aorta-right pulmonary artery anastomosis (Waterston shunt) may impose severe restrictions in the operability of these patients because of severe distortion, narrowing, and scarring of the proximal right pulmonary artery. A modification of the Fontan operation is described in which repair is facilitated by transecting the right pulmonary artery at the site of anastomosis, trimming away the distorted narrowed segment, and anastomosing the resulting cuff to the lateral aspect of the superior vena cava. The right atrial appendage is anastomosed to the rudimentary right ventricle or the main pulmonary artery directly or via a tubular Dacron graft. The technique has been applied successfully in four consecutive cases.

Physiological rationale for a bidirectional cavopulmonary shunt. A versatile complement to the Fontan principle.

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The original Fontan procedure included a classic superior vena cava-to-right pulmonary artery (Glenn) shunt. Subsequent experience demonstrated that this anastomosis was not essential and was an unnecessary commitment of the larger right pulmonary circulation to the smaller blood volume of the superior vena caval return. With application of the Fontan principle to more complex cardiac malformations, there has been a reconsideration of possible benefits of a cavopulmonary shunt in selected patients. A modified shunt from the divided end of the superior vena cava to the side of the undivided right pulmonary artery utilized in 21 patients is described. This shunt is designed to allow bidirectional pulmonary arterial distribution of both superior vena caval inflow and right atrial outflow after completion of the Fontan procedure. Twelve patients had the bidirectional shunt performed prior to a Fontan operation; five of these had a subsequent atriopulmonary connection and seven await operation. Eight patients had construction of this shunt at the time of their Fontan procedure. One patient had a bidirectional shunt constructed following atriopulmonary anastomosis to help relieve right atrial outflow obstruction. Two patients with univentricular heart undergoing simultaneous Fontan procedure and a bidirectional shunt died while in the hospital. The remaining 19 patients have been followed up for 2 months to 9 years with one late sudden death at 9 years. There have been no bidirectional cavopulmonary shunt failures, stenoses, kinks, or recognized pulmonary arteriovenous malformations. Postoperatively, eight patients had assessment of pulmonary distribution of shunt blood flow by angiography. Seven of these patients were also evaluated by radionuclide angiography. Superior vena caval blood flow via the bidirectional cavopulmonary shunt tended to be greater to the right lung, but bilateral pulmonary flow was documented in all but one patient. After Fontan operation, six of seven patients tested also demonstrated bilateral distribution of atriopulmonary flow. We concluded from our experience that this modified shunt provides excellent relief of cyanosis, allows bidirectional pulmonary distribution of both superior vena caval return and also the right atrial blood flow after atriopulmonary connection, and may be done before, with, or after a Fontan procedure and is compatible with all currently recommended modifications. Perioperative hemodynamic adjustments to the Fontan procedure may be improved by reducing atrial volume, and this may also be of potential benefit in the long-term adaptation to Fontan physiology by minimizing atrial distention.